

ECSE-4040

Digital Electronics

Lecture 18

Bipolar Digital Circuits

Profs. Mona Hella and T. Paul Chow
Rensselaer Polytechnic Institute
Troy, NY 12180-3590

References: Appendix B: HJS

Chapters 5 and 17, Neamen

<https://lms.rpi.edu>

Lecture Outline

- BJT SPICE Representation
- VTC of a BJT Inverter
- DC Analysis
- Example of digital circuits

High Speed Logic Families

Emitter-Coupled Logic (ECL)

- Bipolar
- III-V FET
- BiCMOS



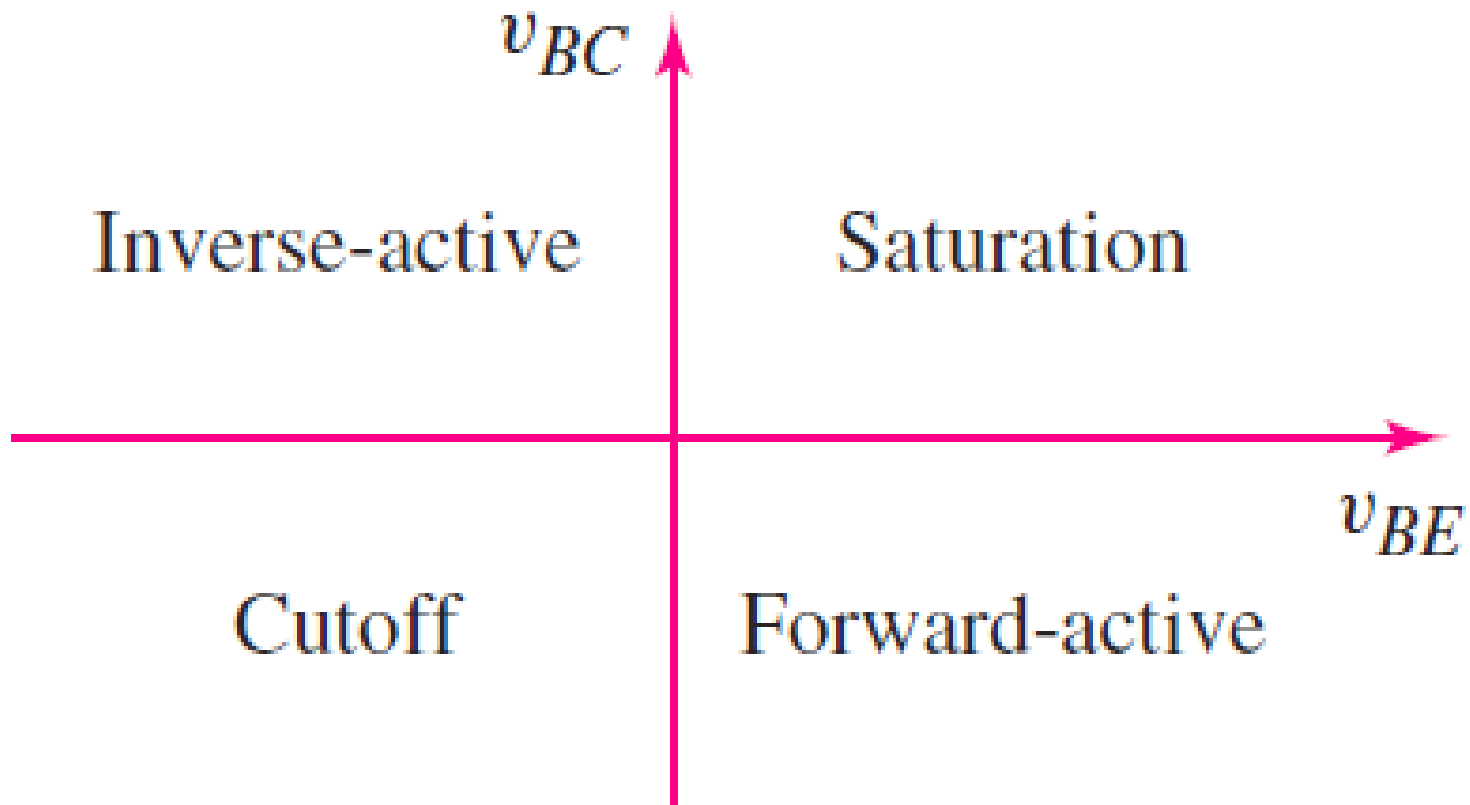
Current-Mode Logic (CML)

- ☐ Bipolar
- ☐ MOS
- ☐ BiCMOS

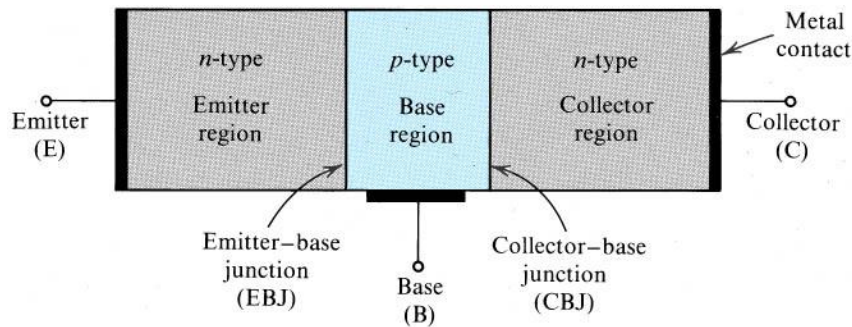


**Some high-speed logic families use differential topologies

Summary of BJT Modes of Operation (NPN)



Summary of Modes of operation-Simplified Analysis



We have two junctions $\rightarrow$ 4 possible models of operation

1. Cut-off

B-E & B-C junctions are reverse biased.

$I_B = I_C = I_E = 0$ (ideal case, the actual value equals the reverse saturation current of the junction)

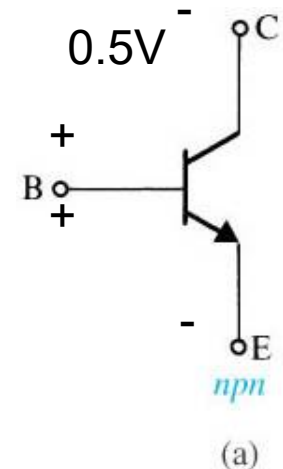
1. Saturation

$$V_{BE}(\text{Sat}) = 0.8V$$

B-E & B-C junctions are forward biased.

$$V_{BE} = 0.7, V_{BC} = 0.5, V_{CE} = 0.2V$$

$$I_E = I_C + I_B$$



Summary of Modes of operation

3. Forward Active

B-E junction is forward biased

B-C junction is reversed biased

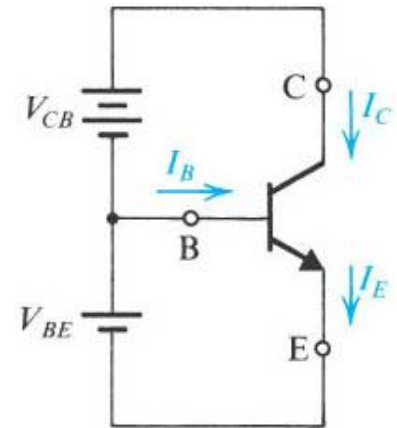
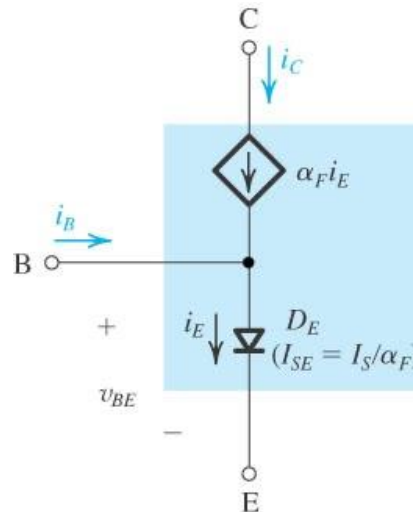
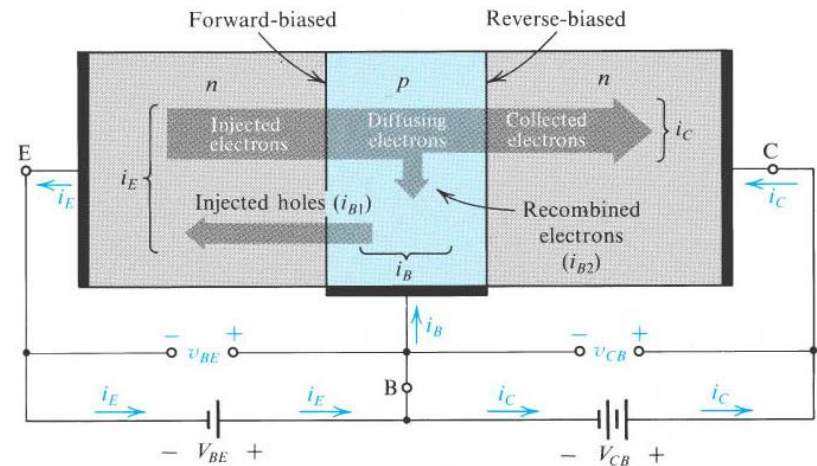
Most electrons injected at the emitter go across the base to the collector

$I_C = \alpha_F I_E$ (α_F is the fraction of electrons going to the collector ~ 1 (0.96-0.99))

$$I_E = I_C + I_B$$

$$\frac{I_C}{I_B} = \beta_F, \frac{I_C}{I_E} = \alpha_F$$

$$\alpha_F = \frac{\beta_F}{\beta_F + 1}, \beta_F = \frac{\alpha_F}{1 - \alpha_F}$$



If I_S is not given

RPI

Summary of Modes of operation

4. Reverse Active

B-E junction is Reverse biased

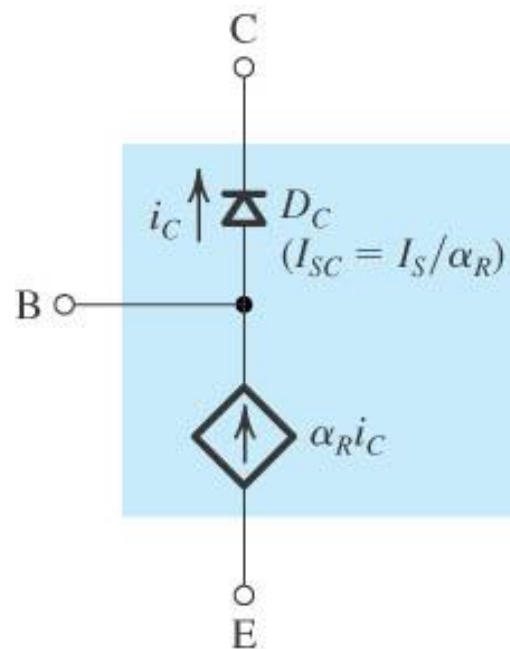
B-C junction is forward biased

$I_C = \alpha_R I_E$ (fraction of electrons from collector that reach the emitter $\ll 1$)

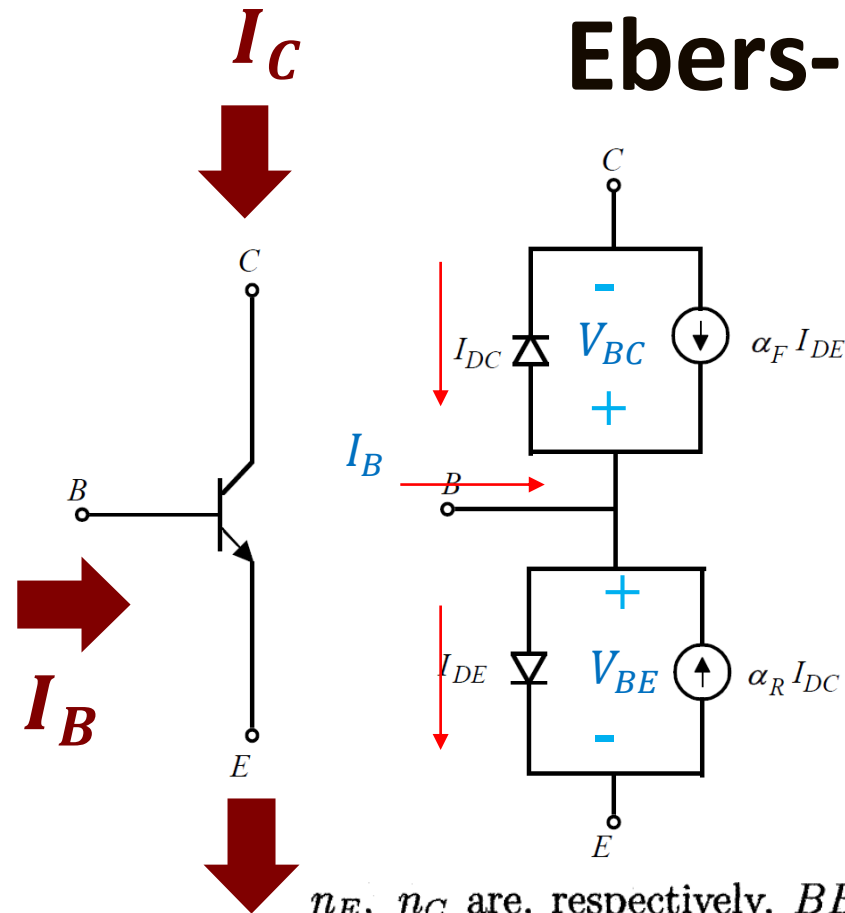
$$\frac{I_C}{I_B} = \beta_R, \frac{I_C}{I_E} = \alpha_R$$

$$\alpha_R = \frac{\beta_R}{\beta_R + 1}$$

$$\beta_R \ll \ll \beta_F$$



Ebers-Moll Model



$$I_{DE} = I_{ES} \left(e^{V_{BE}/n_E V_T} - 1 \right)$$

$$I_{DC} = I_{CS} \left(e^{V_{BC}/n_C V_T} - 1 \right)$$

n_E, n_C are, respectively, BE and BC junction diode ideality factors (emission constants).

I_{ES}, I_{CS} are respective reverse saturation currents for BE and BC junctions. Typical values: $I_{ES}, I_{CS} \sim 10^{-15} a$.

α_F, α_R are respective *forward and reverse transport fractions*. $E \rightarrow C$, forward injection; $C \rightarrow E$, reverse injection. Typical values: $\alpha_F \sim 0.99, \alpha_R \sim 0.66$.

Ebers-Moll Model

Kirchoff's current law (KCL) provides BJT terminal currents:

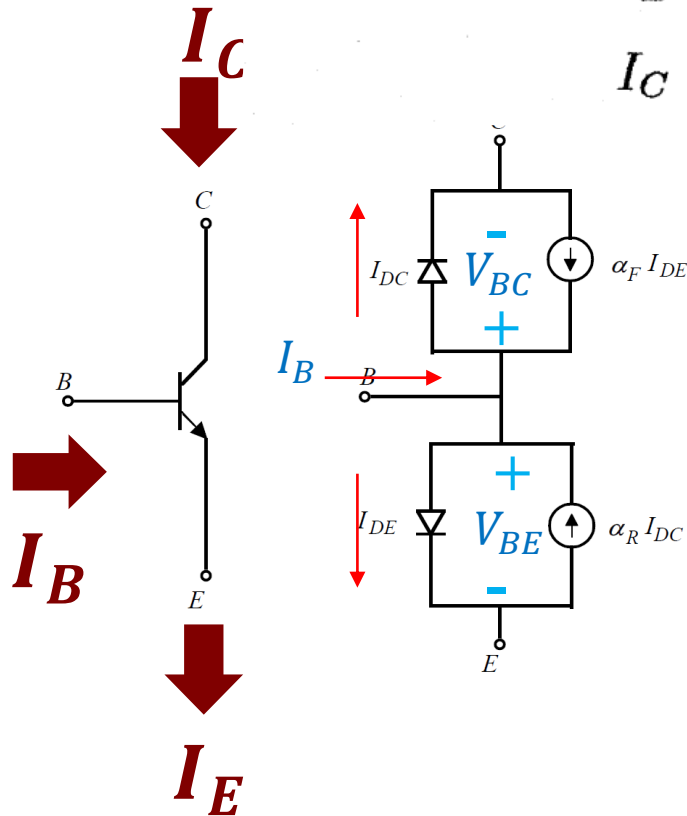
$$I_E = I_{ES} \left(e^{V_{BE}/n_E V_T} - 1 \right) - \alpha_R I_{CS} \left(e^{V_{BC}/n_C V_T} - 1 \right)$$

$$I_C = \alpha_F I_{ES} \left(e^{V_{BE}/n_E V_T} - 1 \right) - I_{CS} \left(e^{V_{BC}/n_C V_T} - 1 \right)$$

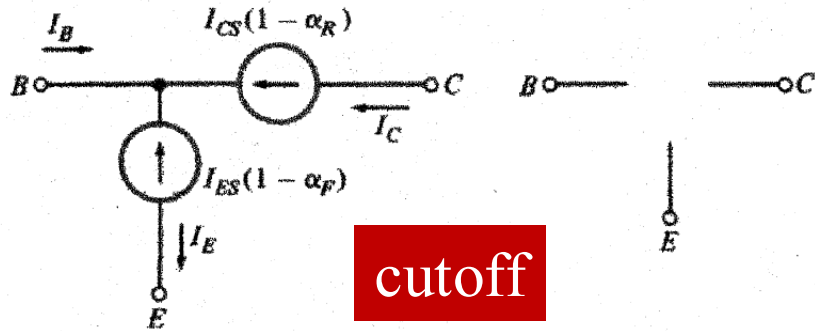
$$I_B = I_E - I_C$$

Based on the E-M model, one can also prove *reciprocity*:

$$\alpha_F I_{ES} = \alpha_R I_{CS}.$$

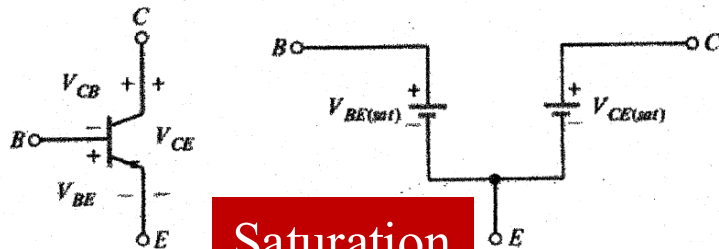


E-M Model & Regions of Operation



cutoff

With and w/o junction leakage currents

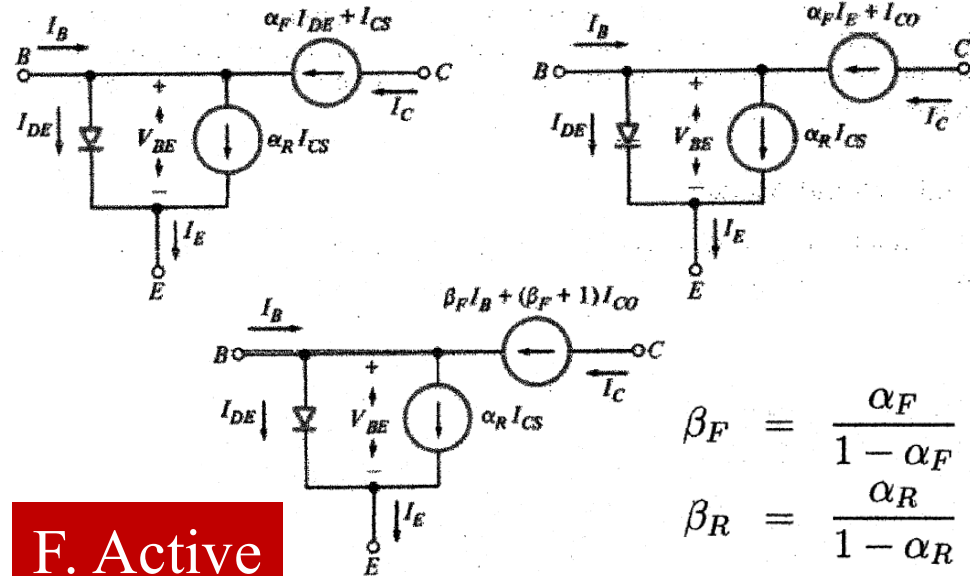


Saturation

$$V_{BE}(sat) = 0.8v, \quad V_{CE}(sat) = 0.2v.$$

$$\beta_F I_B > I_C.$$

$I_C/I_B \doteq \beta(sat)$ in SAT is called the *forced beta*



F. Active

$$\beta_F = \frac{\alpha_F}{1 - \alpha_F}$$

$$\beta_R = \frac{\alpha_R}{1 - \alpha_R}$$

R. Active

Same as FA case, except $E \leftrightarrow C$ on all subscripts

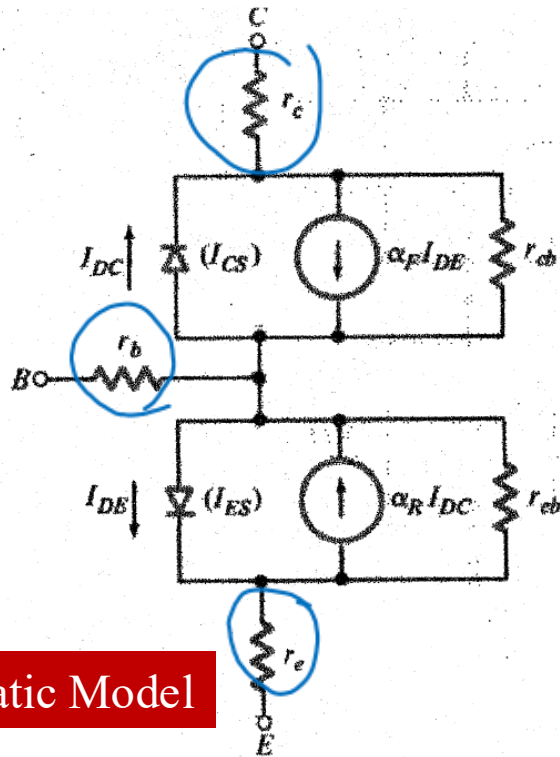
$$\beta_F \leftrightarrow \beta_R, \quad \alpha_F \leftrightarrow \alpha_R$$

RPI

BJT SPICE

Diffusion charge

Depletion charge



Static Model

$$Q_{BE} = \tau_F I_S (e^{V_{BE}/n_E V_T} - 1) + C_{jE0} \int_0^{V_{BE}} dV \left(1 - \frac{V}{\phi_E}\right)^{-m_E}$$

$$Q_{BC} = \tau_R I_S (e^{V_{BC}/n_C V_T} - 1) + C_{jC0} \int_0^{V_{BC}} dV \left(1 - \frac{V}{\phi_C}\right)^{-m_C}$$

Nonlinear charge storage capacitance → transient Model

$$I_E = \frac{I_S}{\alpha_F} (e^{V_{BE}/n_E V_T} - 1) - I_S (e^{V_{BC}/n_C V_T} - 1)$$

$$I_C = I_S (e^{V_{BE}/n_E V_T} - 1) - \frac{I_S}{\alpha_R} (e^{V_{BC}/n_C V_T} - 1)$$

$$I_s = \alpha_F I_{ES} = \alpha_R I_{CS}$$

Go to Review Slides to review pn
junction capacitances

BJT SPICE

BJT syntax

Q<name> <collector node> <base node> <emitter node> [<substrate node>] <modelname>

+ [<area>]

.MODEL <modelname> NPN(BF=<value> BR=<value> IS=<value> CJE=<value>

+ CJC=<value> VJE=<value> VJC=<value> VAF=<value> VAR=<value>

+ NF=<value> NR=<value>)

BJT Parameters

BF Forward active current gain

VJE Base-emitter built-in potential

BR Reverse active current gain

VJC Base-collector built-in potential

IS Transport saturation current

VAF Forward mode Early voltage

CJE Base-emitter zero-bias

VAR Reverse mode Early voltage

junction capacitance

NF Forward mode ideality factor

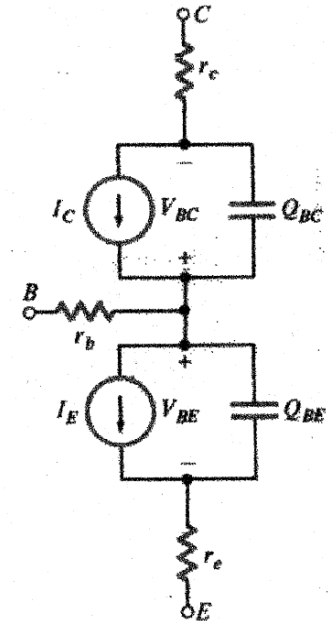
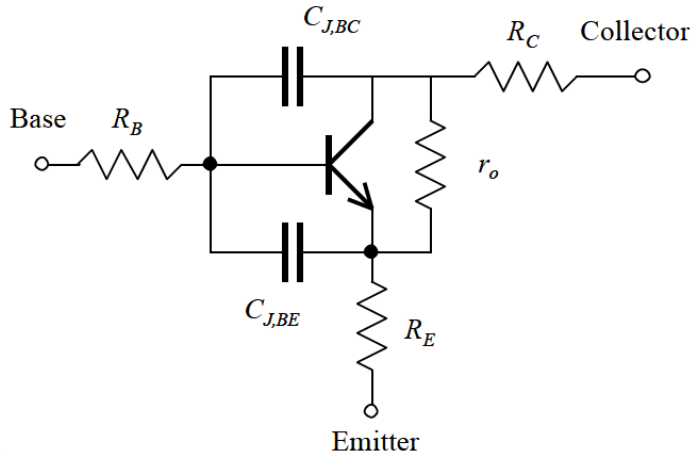
CJC Base-collector zero-bias

NR Reverse mode ideality factor

Junction capacitance



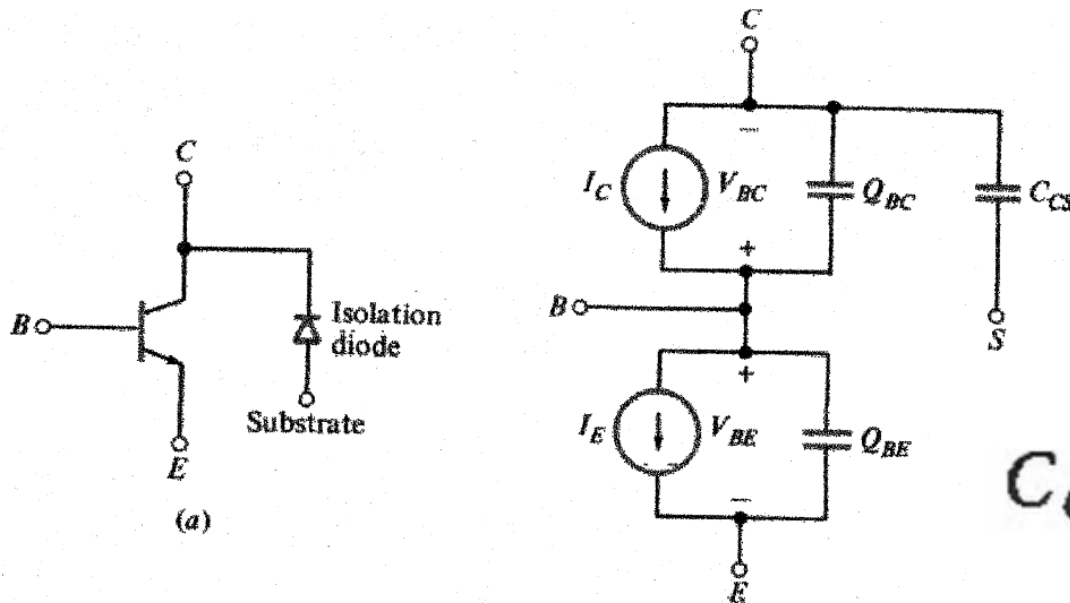
BJT SPICE Model



$$C_{BE} = \frac{dQ_{BE}}{dV_{BE}} = \frac{\tau_F I_S}{V_T} e^{V_{BE}/V_T} + \frac{C_{je0}}{[1 - (V_{BE}/\phi_e)]^{m_e}}$$

$$C_{BC} = \frac{dQ_{BC}}{dV_{BC}} = \frac{\tau_R I_S}{V_T} e^{V_{BC}/V_T} + \frac{C_{jc0}}{[1 - (V_{BC}/\phi_c)]^{m_c}}$$

BJT(IC) SPICE Model



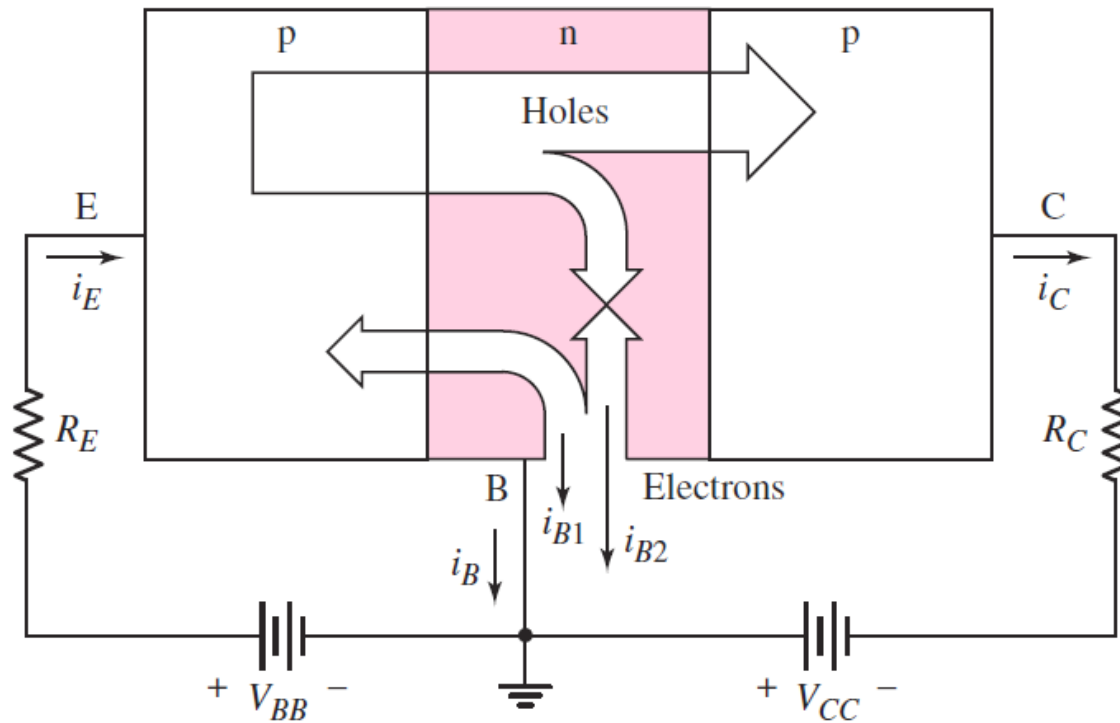
$$C_{CS} = \frac{C_{jso}}{[1 - (V_{sc}/\phi_s)]^{m_s}}$$

For ICs, an isolation diode has to be considered. The diode is always reverse biased $\rightarrow$ SPICE considers only depletion capacitance

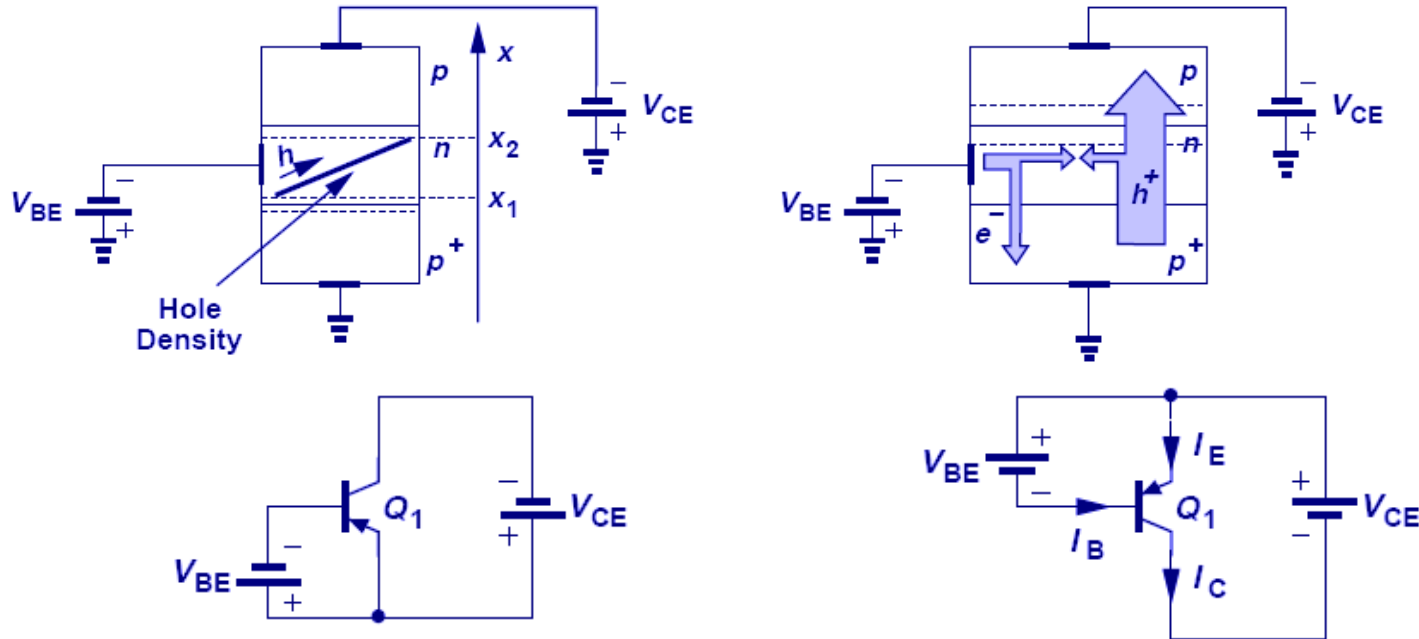
BJT SPICE

SPICE Parameter	Model Constant	Brief Description	Typical Value
BF	β	Transistor current gain in forward direction	100
ISS	I_{CS}	Transport saturation current in forward direction	2×10^{-18} A
VAF	V_A	Early voltage in forward direction	50 V
RB	r_b	Series base resistance	500 Ω
RE	R_E	Series emitter resistance	30 Ω
CJE	C_{je0}	Base-emitter depletion capacitance at 0 V	0.015 pF
CJC	C_{je0}	Base-collector depletion capacitance at 0 V	0.018 pF
CJS	C_{je0}	Collector-substrate depletion capacitance at 0 V	0.040 pF
MJE	m_e	Base-emitter junction exponent (grading factor)	0.30
MJC	m_c	Base-collector junction exponent (grading factor)	0.35
MJS	m_s	Collector-substrate junction exponent (grading factor)	0.29
TF	τ_F	Forward-base transit time	12 ps
TR	τ_R	Reverse-base transit time	4 ns

Bipolar PNP Transistor



PNP Transistor



- With the polarities of emitter, collector, and base reversed, a PNP transistor is formed.
- All the principles that applied to NPN's also apply to PNP's, with the exception that emitter is at a higher potential than base and base at a higher potential than collector.

A Comparison between NPN and PNP

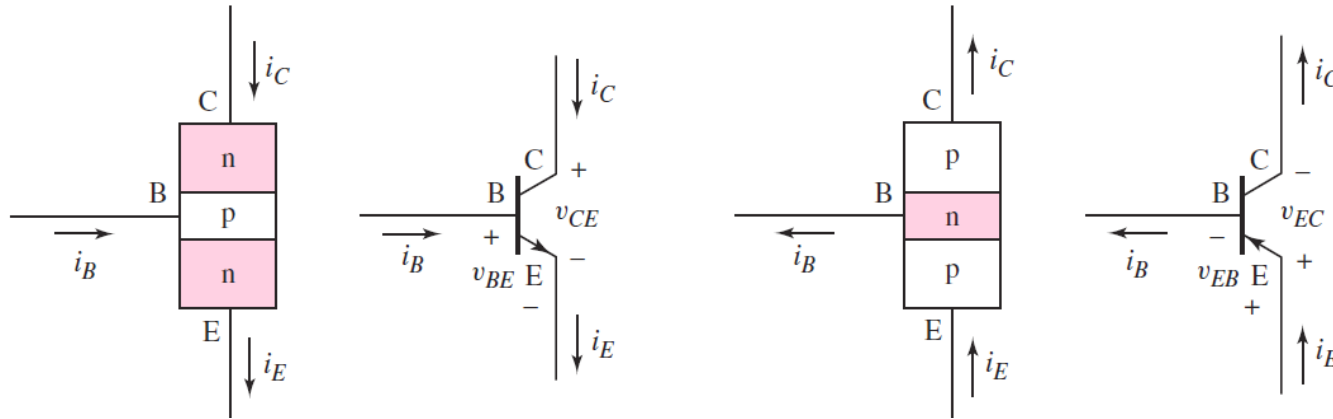
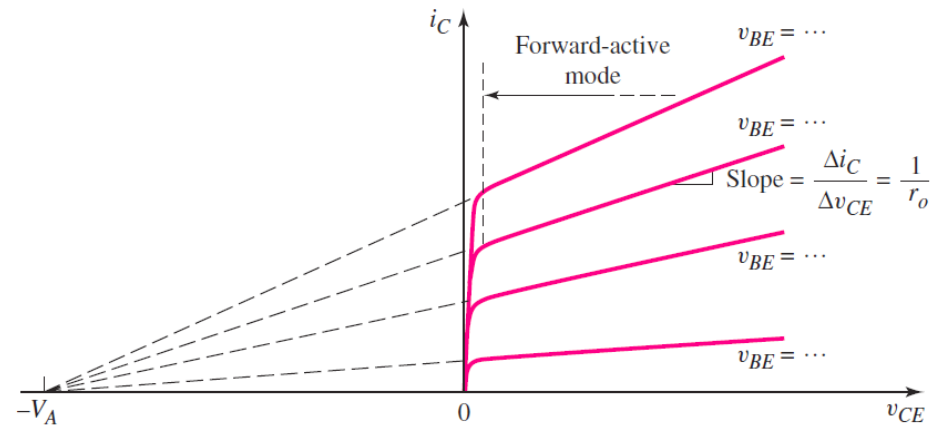
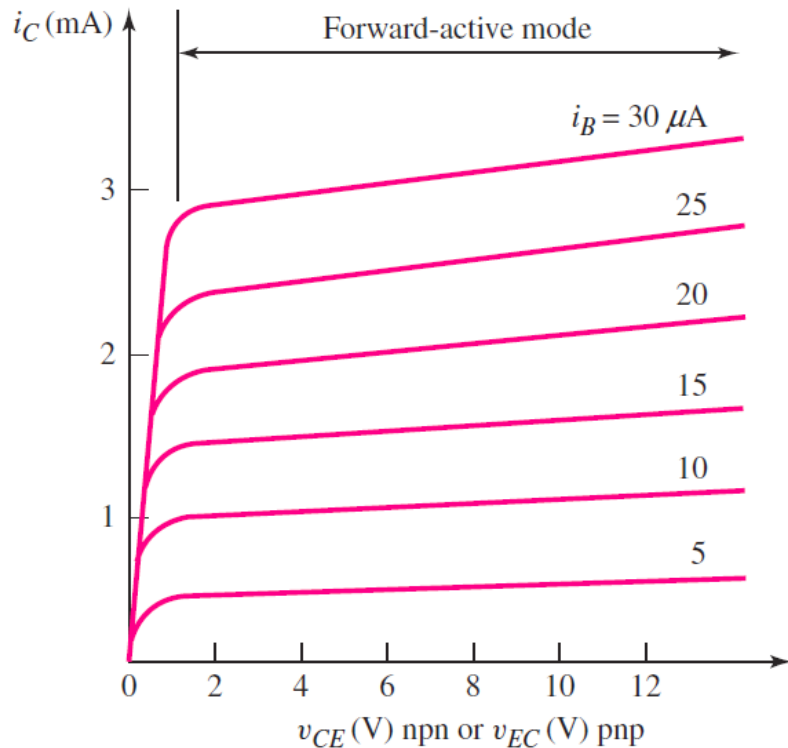


Table 5.1

Summary of the bipolar current-voltage relationships in the active region

npn	pnp
$i_C = I_S e^{v_{BE}/V_T}$	$i_C = I_S e^{v_{EB}/V_T}$
$i_E = \frac{i_C}{\alpha} = \frac{I_S}{\alpha} e^{v_{BE}/V_T}$	$i_E = \frac{i_C}{\alpha} = \frac{I_S}{\alpha} e^{v_{EB}/V_T}$
$i_B = \frac{i_C}{\beta} = \frac{I_S}{\beta} e^{v_{BE}/V_T}$	$i_B = \frac{i_C}{\beta} = \frac{I_S}{\beta} e^{v_{EB}/V_T}$
For both transistors	
$i_E = i_C + i_B$	$i_C = \beta i_B$
$i_E = (1 + \beta) i_B$	$i_C = \alpha i_E = \left(\frac{\beta}{1 + \beta} \right) i_E$
$\alpha = \frac{\beta}{1 + \beta}$	$\beta = \frac{\alpha}{1 - \alpha}$

More on Early Effect

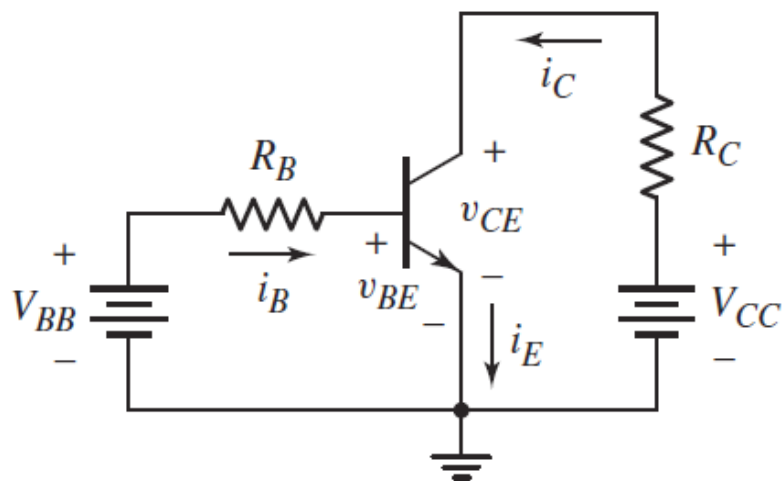


$$r_o \cong \frac{V_A}{I_C}$$

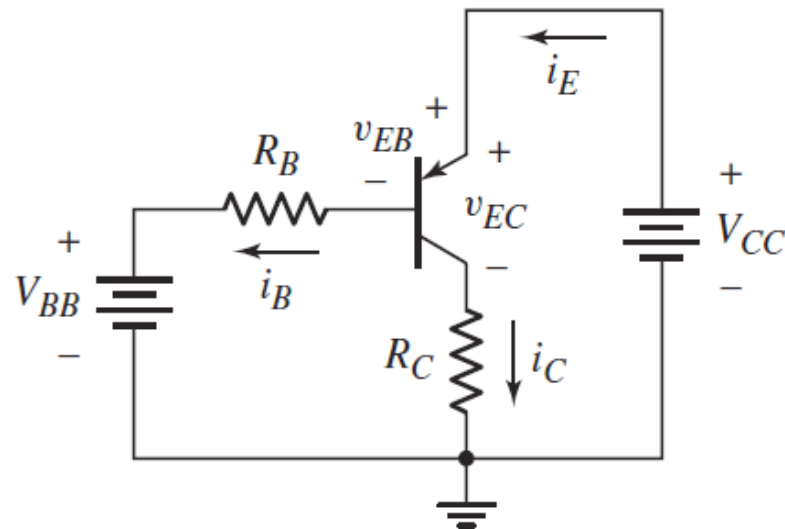
Test Your Understanding

TYU 5.4 The output resistance of a bipolar transistor is $r_o = 225 \text{ k}\Omega$ at $I_C = 0.8 \text{ mA}$.
(a) Determine the Early voltage. (b) Using the results of part (a), find r_o at
(i) $I_C = 0.08 \text{ mA}$ and (ii) $I_C = 8 \text{ mA}$. (Ans. (a) $V_A = 180 \text{ V}$; (b) (i) $r_o = 2.25 \text{ M}\Omega$,
(ii) $r_o = 22.5 \text{ k}\Omega$)

NPN & PNP Biasing

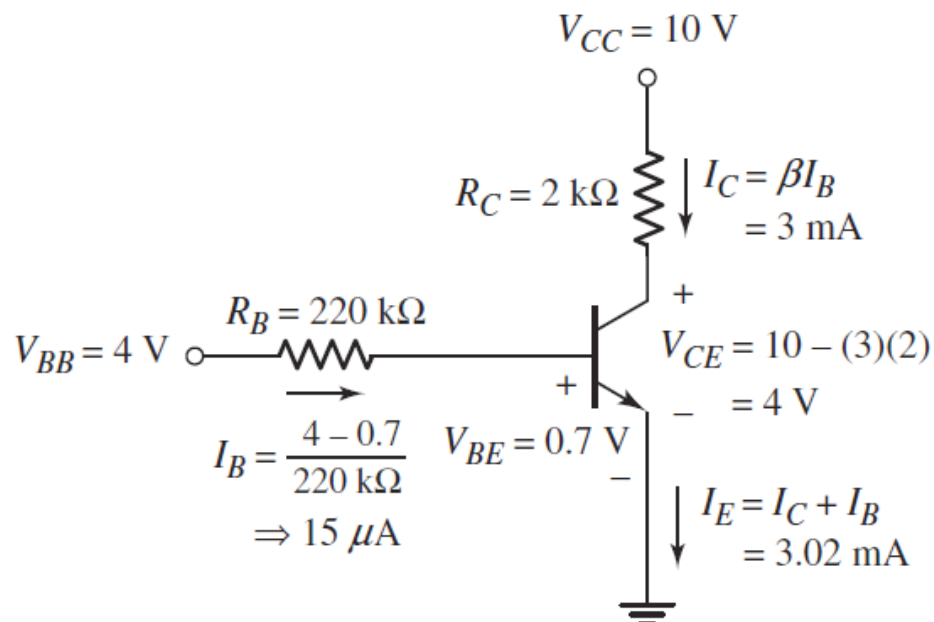
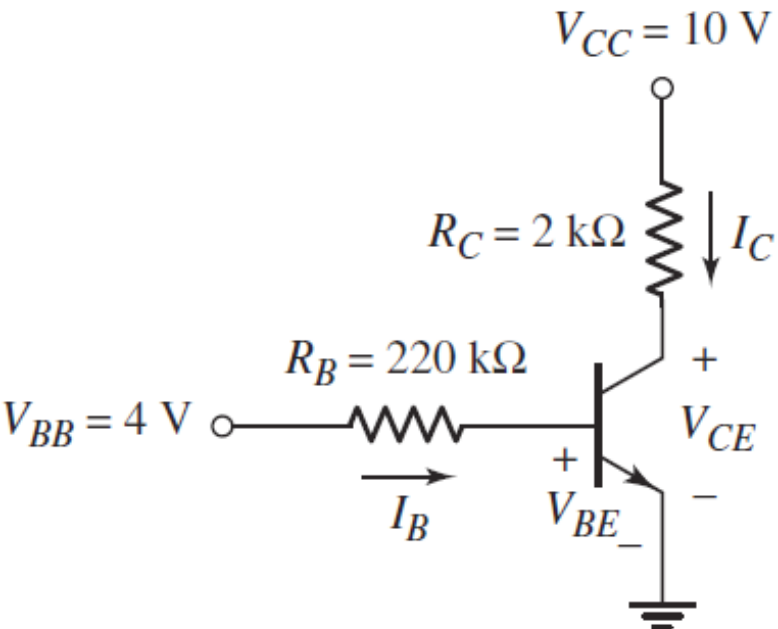


NPN: Collector is at a higher potential than both the base and Emitter.



PNP: Emitter is at a higher potential than both the base and collector.

DC-Analysis Examples

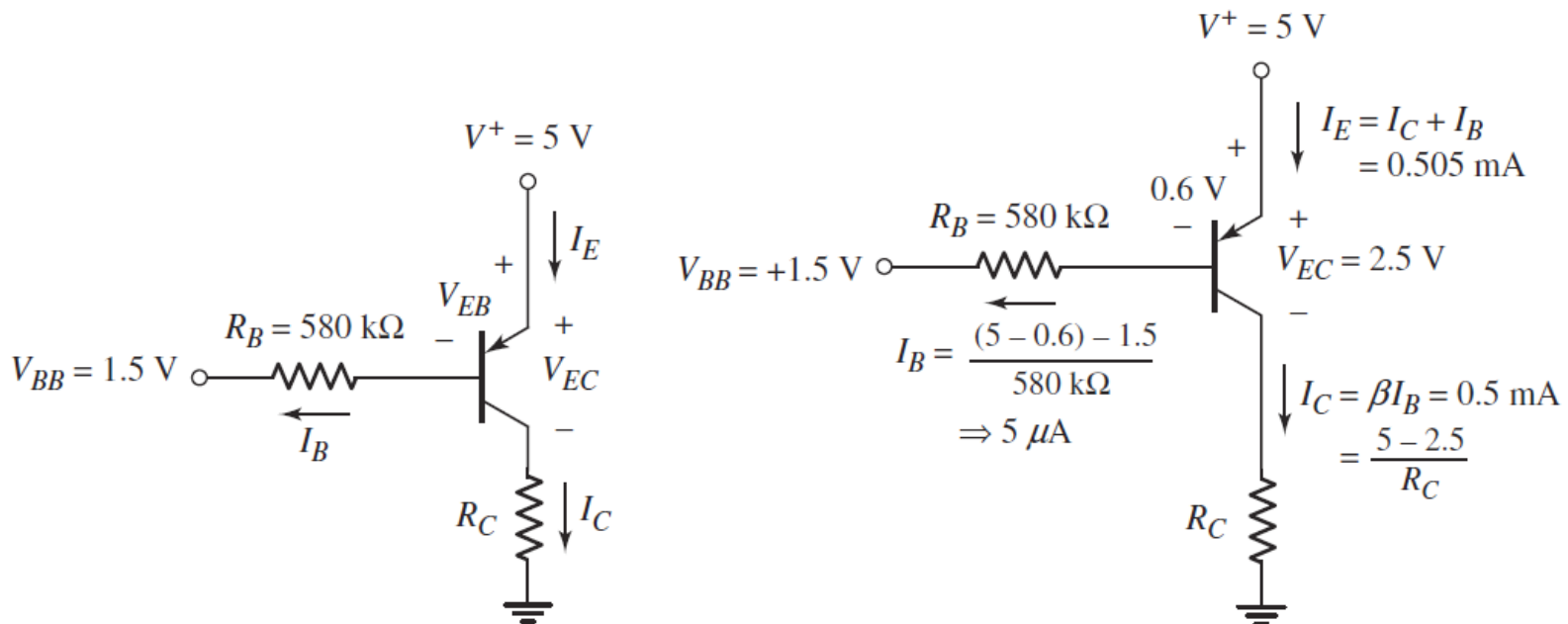


What is the power dissipated in the transistor?

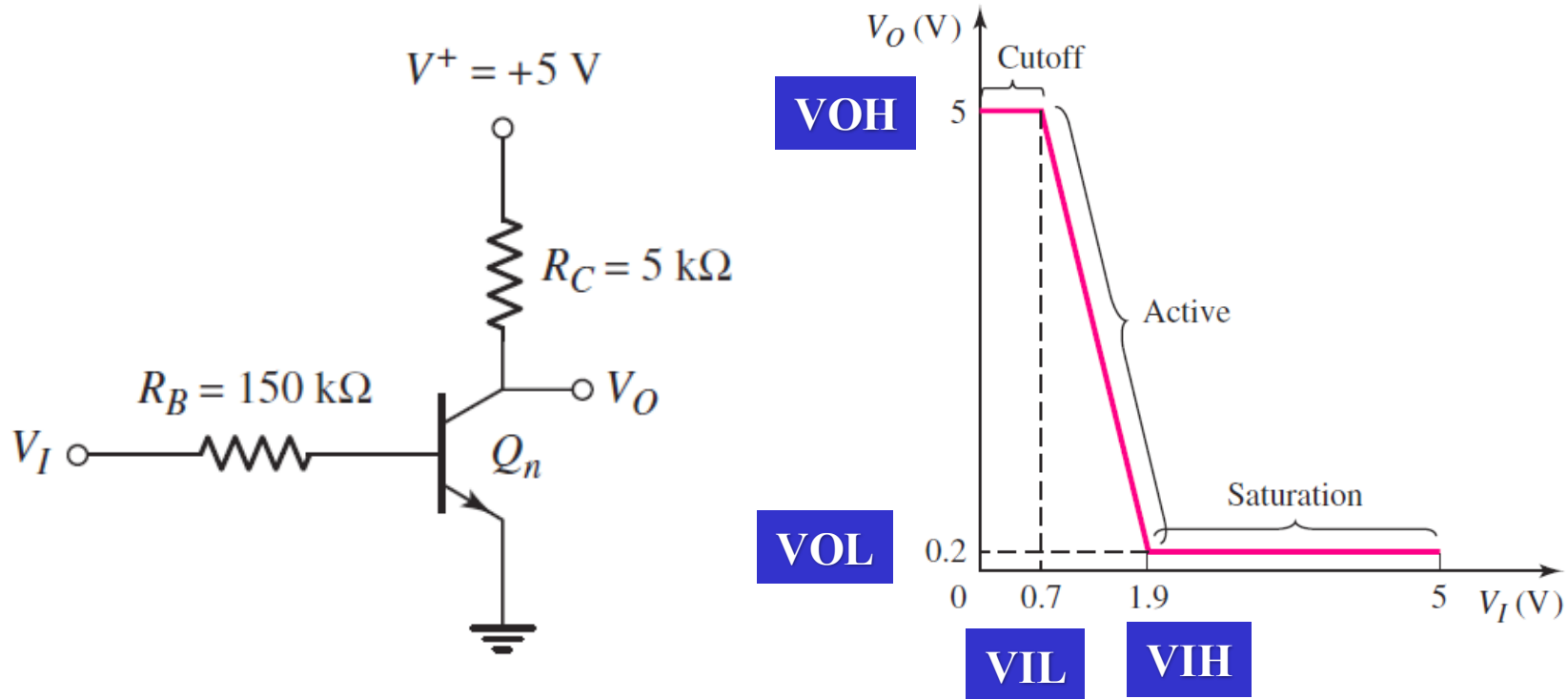
$$P_T = I_B V_{BE(\text{on})} + I_C V_{CE} = (0.015)(0.7) + (3)(4) \cong I_C V_{CE}$$

PNP DC Analysis Example

For the circuit shown in Figure 5.22(a), the parameters are: $V_{BB} = 1.5$ V, $R_B = 580$ k Ω , $V^+ = 5$ V, $V_{EB(\text{on})} = 0.6$ V, and $\beta = 100$. Find I_B , I_C , I_E , and R_C such that $V_{EC} = (\frac{1}{2})V^+$. **Use $V_{EB}=0.6$ V**

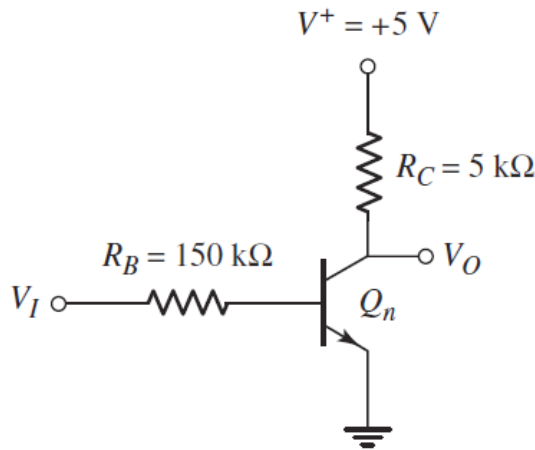


Voltage Transfer C/Cs (VTC)-NPN



Assume npn transistor parameters of $V_{BE}(\text{on}) = 0.7 \text{ V}$, $\beta = 120$, $V_{CE}(\text{sat}) = 0.2 \text{ V}$, and $V_A = \infty$,

Voltage Transfer C/Cs (VTC)-NPN



1. V_{OH} , which is equivalent to V_{CE} with the transistor at the edge of the cutoff region, that is, V_{CC} . For the given example it is 5 V. Hence,

$$V_{OH} = V_{CC}$$

However, V_{OH} is usually measured with $V_{in} = V_{OL}$

2. V_{OL} , which is equivalent to V_{CE} with the transistor at the edge of the saturation region, that is, $V_{CE(sat)}$. In the example it is 0.1 V. Thus,

$$V_{OL} = V_{CE(sat)}$$

V_{OL} is commonly measured with $V_{in} = V_{OH}$

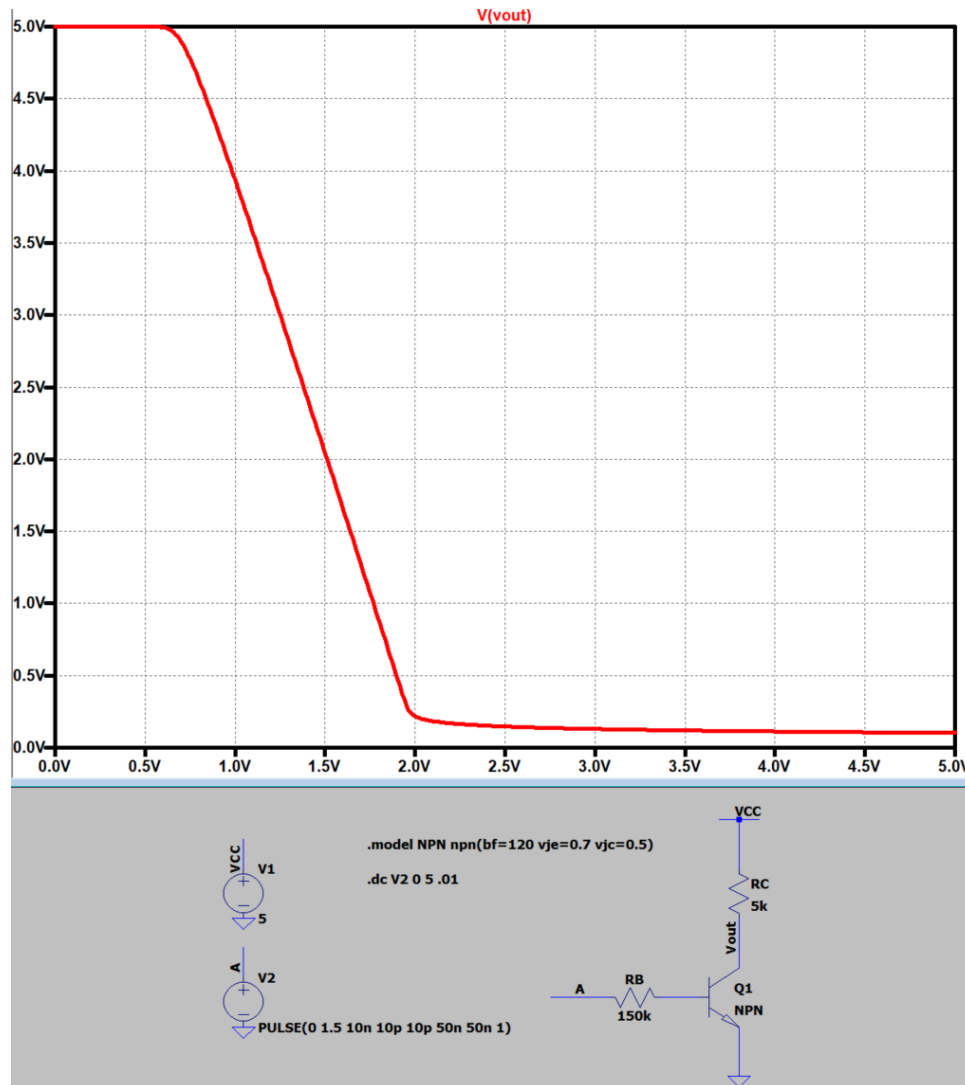
3. V_{IL} , which is the input voltage just sufficient to turn the transistor on. In this example it is $V_{BE(on)}$, which is given as 0.7 V. Therefore,

$$V_{IL} = V_{BE(on)}$$

4. V_{IH} , which is the input voltage just sufficient to saturate the transistor. Now with the transistor just at the edge of saturation, $I_C = I_{C(EOS)}$, where

$$V_{IH} = V_{BE(sat)} + \frac{R_B}{R_C} \frac{V_{CC} - V_{CE(sat)}}{\beta_F}$$

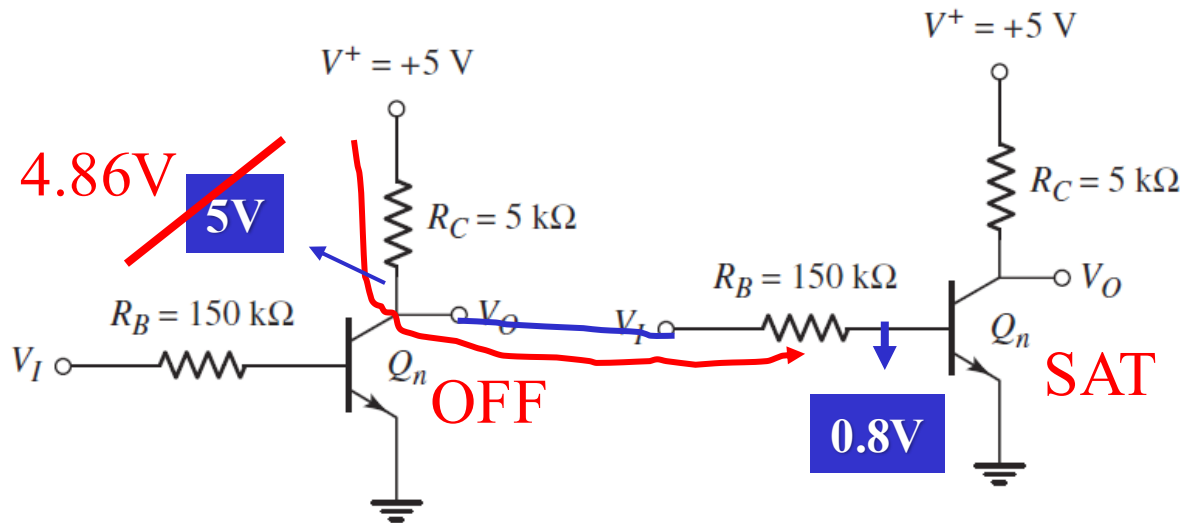
Voltage Transfer C/Cs (VTC)-NPN



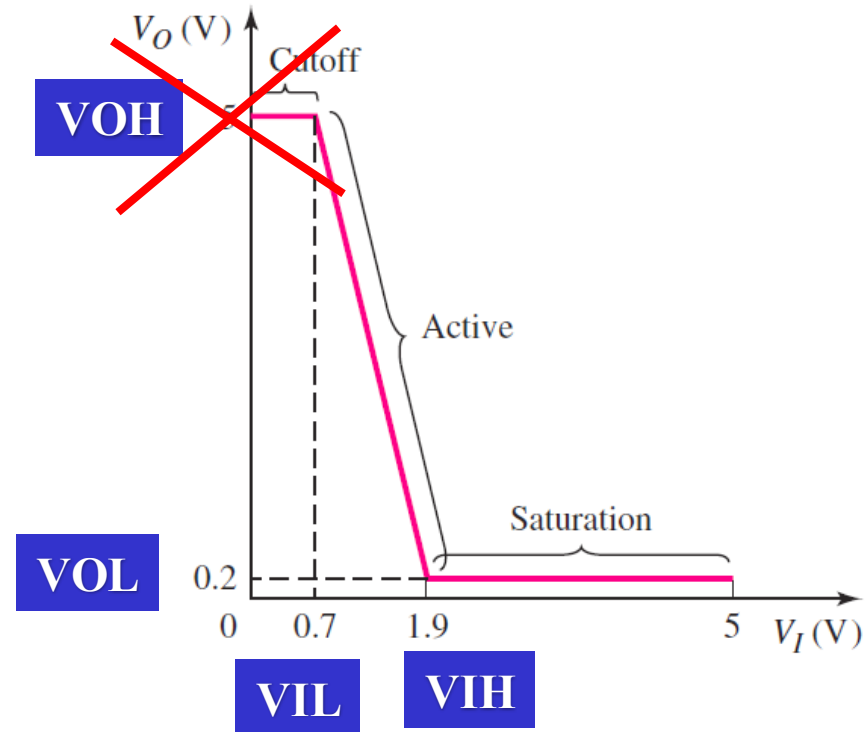
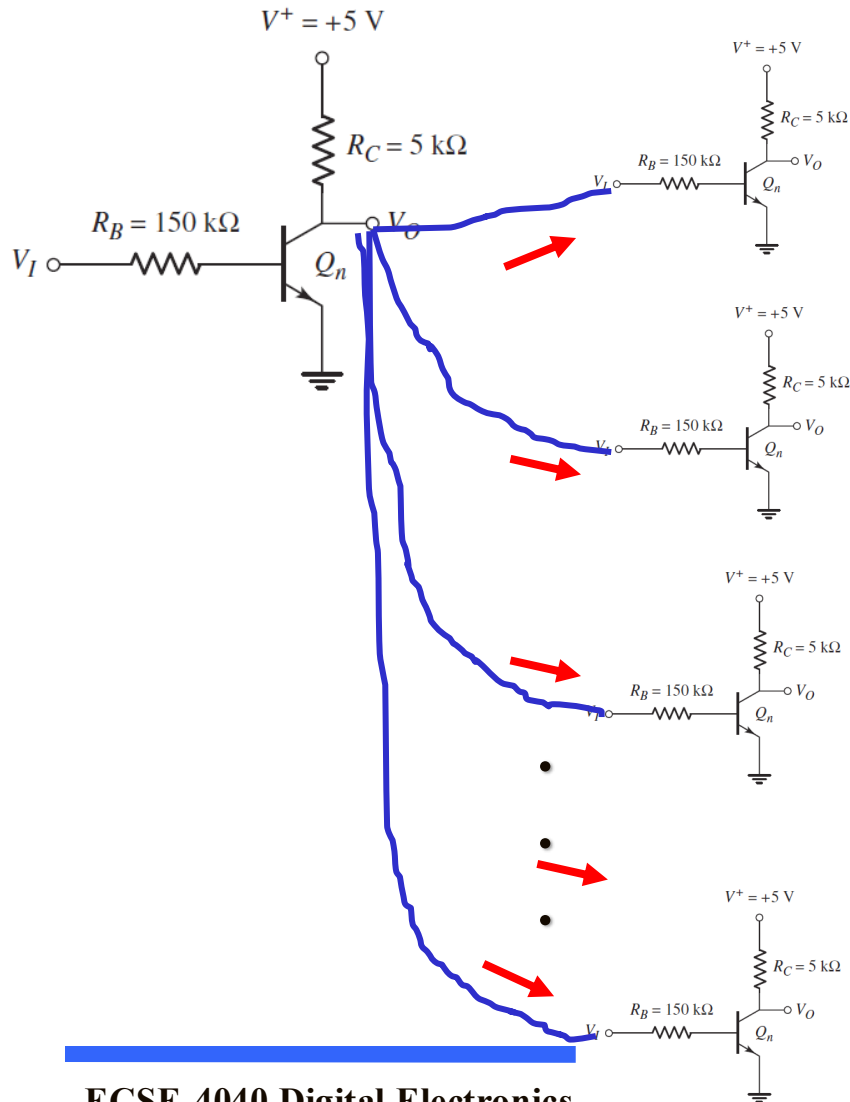
Voltage Transfer C/Cs (VTC)-NPN

Assume npn transistor parameters of $V_{BE}(\text{on}) = 0.7 \text{ V}$, $\beta = 120$,
 $V_{CE}(\text{sat}) = 0.2 \text{ V}$, and $V_{BE}(\text{SAT}) = 0.8$

$$V_{out} = (V_{CC} - 0.8) \frac{R_B}{R_B + R_C} + 0.8 =$$

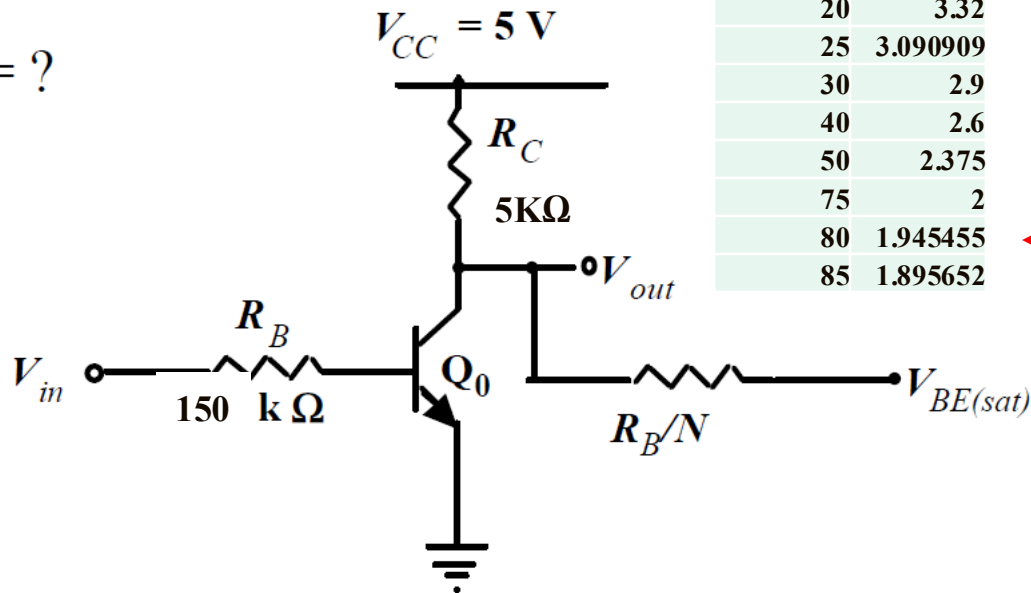


Inverter Fanout



NPN Inverter Fanout

$V_{OH} = ?$

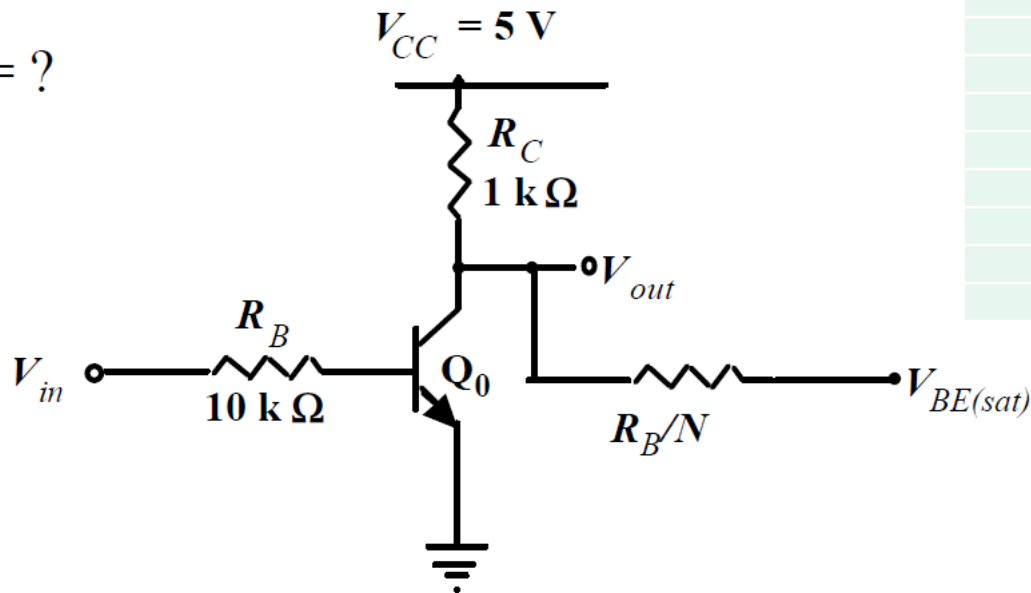


0	5
1	4.864516
2	4.7375
4	4.505882
10	3.95
15	3.6
20	3.32
25	3.090909
30	2.9
40	2.6
50	2.375
75	2
80	1.945455
85	1.895652

$$V_{out} = (V_{CC} - 0.8) \frac{R_B / N}{R_B / N + R_C} + 0.8 =$$

NPN Inverter Fanout

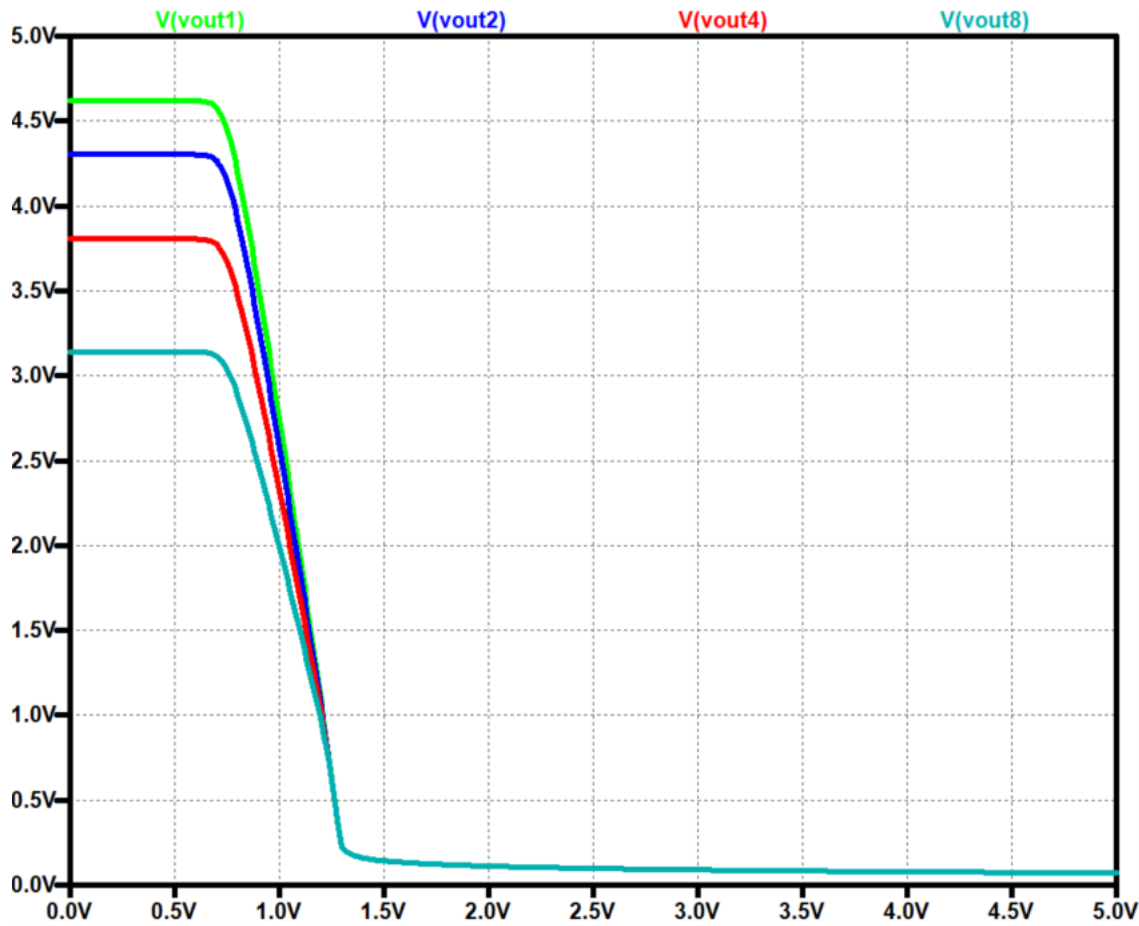
$V_{OH} = ?$



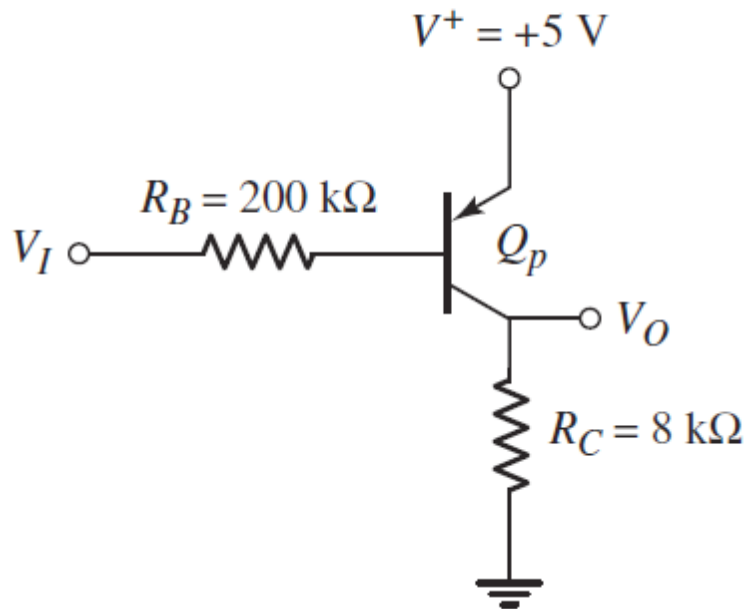
0	5
1	4.618182
2	4.3
4	3.8
10	2.9
15	2.48
20	2.2
25	2
30	1.85
40	1.64
50	1.5
75	1.294118
80	1.266667
85	1.242105

$$V_{out} = (V_{CC} - 0.8) \frac{R_B / N}{R_B / N + R_C} + 0.8 =$$

NPN Inverter Fanout

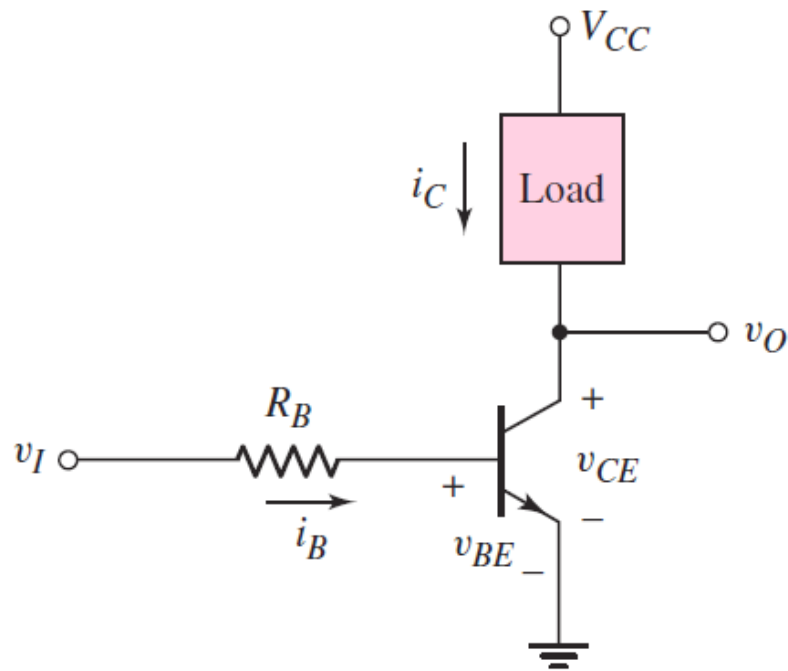


Voltage Transfer C/Cs (VTC)-PNP



Assume pnp transistor parameters of $V_{EB}(\text{on}) = 0.7\text{ V}$, $\beta = 80$, $V_{EC}(\text{sat}) = 0.2\text{ V}$, and $V_A = \infty$.

BJT As a Switch



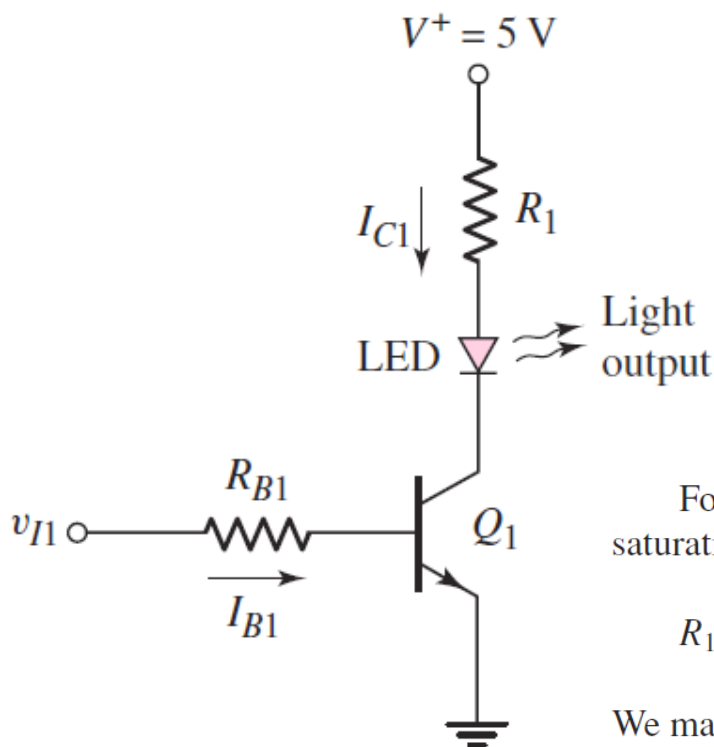
$$i_B \cong \frac{v_I - V_{BE}(\text{on})}{R_B}$$

$$i_C = I_C(\text{sat}) = \frac{V_{CC} - V_{CE}(\text{sat})}{R_C}$$

and

$$v_O = V_{CE}(\text{sat})$$

Example



The transistor in the inverter is used to turn the light-emitting diode (LED) on and off. The required LED current is $I_{C1} = 12$ mA to produce the specified output light. Assume transistor parameters of $\beta = 80$, $V_{BE}(\text{on}) = 0.7$ V, and $V_{CE}(\text{sat}) = 0.2$ V, and assume the diode cut-in voltage is $V_\gamma = 1.5$ V. [Note: LEDs are fabricated with compound semiconductor materials and have a larger cut-in voltage compared to silicon diodes.]

For $v_{I1} = 5$ V, we require $I_{C1} = 12$ mA and want the transistor to be driven into saturation. Then

$$R_1 = \frac{V^+ - (V_\gamma + V_{CE}(\text{sat}))}{I_{C1}} = \frac{5 - (1.5 + 0.2)}{12} \Rightarrow R_1 = 275 \, \Omega$$

We may let $I_{C1}/I_{B1} = 40$. Then $I_{B1} = 12/40 = 0.3$ mA. Now

$$R_{B1} = \frac{v_{I1} - V_{BE}(\text{on})}{I_{B1}} = \frac{5 - 0.7}{0.3} = 14.3 \, \text{k}\Omega$$

The power dissipation in Q_1 is

$$P_1 = I_{B1}V_{BE}(\text{on}) + I_{C1}V_{CE}(\text{sat}) = (0.3)(0.7) + (12)(0.2) = 2.61 \, \text{mW}$$

Digital Gates

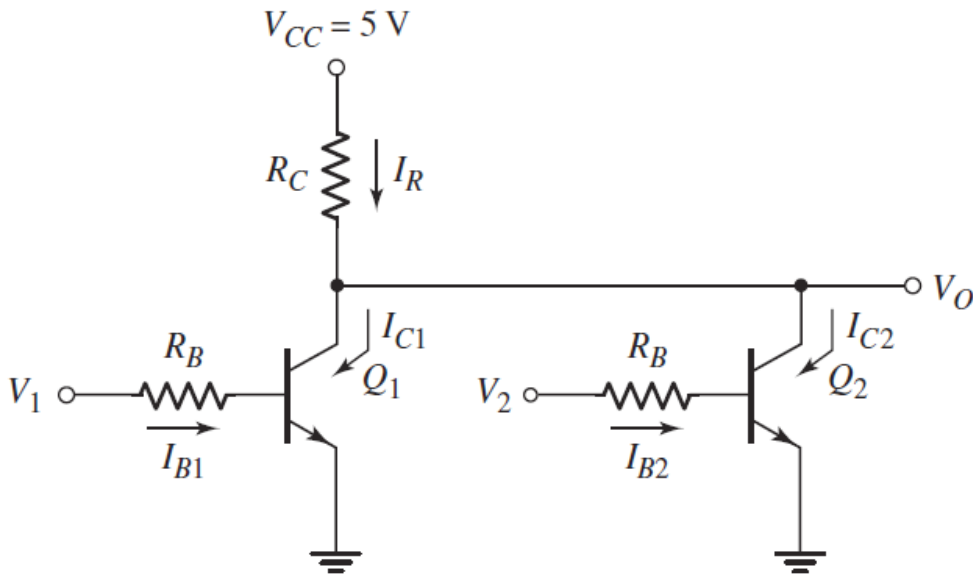


Table 5.2 The bipolar NOR logic circuit response

V_1 (V)	V_2 (V)	V_O (V)
0	0	5
5	0	0.2
0	5	0.2
5	5	0.2

Condition	V_O	I_R	Q_1	Q_2
$V_1 = 0$, $V_2 = 0$	5 V	0	$I_{B1} = I_{C1} = 0$	$I_{B2} = I_{C2} = 0$
$V_1 = 5$ V, $V_2 = 0$	0.2 V	$\frac{5 - 0.2}{1} = 4.8$ mA	$I_{B1} = \frac{5 - 0.7}{20} = 0.215$ mA $I_{C1} = I_R = 4.8$ mA	$I_{B2} = I_{C2} = 0$
$V_1 = 0$, $V_2 = 5$ V	0.2 V	4.8 mA	$I_{B1} = I_{C1} = 0$	$I_{B2} = 0.215$ mA $I_{C2} = I_R = 4.8$ mA
$V_1 = 5$ V, $V_2 = 5$ V	0.2 V	4.8 mA	$I_{B1} = 0.215$ mA $I_{C1} = \frac{I_R}{2} = 2.4$ mA	$I_{B2} = 0.215$ mA $I_{C2} = \frac{I_R}{2} = 2.4$ mA

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